11/20

condition

Q 1. Write a program to check whether a number is positive, negative, or zero

```
num = float(input("Enter a number:"))
if num > 0:
    print("The number is positive")
elif num < 0:
    print("The number is negative")
else:
    print("The number is zero")
```

1

```
Enter a number: 25

The number is positive
```

Q 2. Write a program to find the largest of three numbers.

```
a = int(input("Enter first number:"))
b = int(input("Enter second number:"))
c = int(input("Enter third number:"))
if a >= b and a >= c:
    largest = a
elif b >= a and b >= c:
    largest = b
else:
    largest = c
print("largest number is:", largest)
```

1

```
Enter first number: 80
Enter second number: 40
Enter third number: 50

largest number is: 80
```

Q3. Write a program to check whether a given year is a leap year.

```
year =int(input("Enter a year:"))

if(year % 400 == 0) or (year % 4 == 0 and year % 100 !=0):
    print("leap year")
else:
    print("Not a leap year")
```

1

```
Enter a year: 1990

Not a leap year
```

Q 4. Write a program to check if a character is a vowel or consonant.

```
str = input("Enter a character:").lower()

if str in "aeiou":
```

```
    print("The character is avowel")
elif str.isalpha():
    print("The character is a consonant")
else:
    print(" please enter a valid character")

Enter a character: h

The character is a consonant
```

Q 5. Write a program to calculate the electricity bill:

- First 100 units → ₹5/unit
- Next 100 units → ₹7/unit
- Above 200 units → ₹10/unit

```
units = int(input("Enter units :"))
if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = ( units * 7) - 200
else:
    bill = (units * 10) - 800
    print("Total bill:", bill)

Enter units : 400

Total bill: 3200
```

Loops (for / while)

1. Print all even numbers from 1 to 100.

```
for i in range(1, 101):
    if i % 2 == 0:
        print(i)

2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
```

1

```
44
46
48
50
52
54
56
58
60
62
64
66
68
70
72
74
76
78
80
82
84
86
88
90
92
94
96
98
100
```

2. Find the factorial of a given number using a loop.

```
num = int(input("Enter a number:"))
fact =1

for i in range(1, n + 1):
    fact = fact *i
    print("factorial of", n, "is:", fact)

Enter a number: 8

factorial of 5 is: 1
factorial of 5 is: 2
factorial of 5 is: 6
factorial of 5 is: 24
factorial of 5 is: 120
```

1

3. Print the Fibonacci series up to n terms.

```
n = int(input("Enter a number:"))
a = 0
b = 1
i = 0
print("fibonacci series:")
while i < n:
    print(a, end=" ")
    c = a + b
    a = b
```

a,b=b,a+b.

1

```
        b = c
        i = i + 1

    Enter a number: 6

    fibonacci series:
    0 1 1 2 3 5
```

4. Find the sum of digits of a given number.

```
    n = input("Enter a number:")
    sum = 0
    for digit in n:
        sum = sum + int( digit)
        print ("sum of digits:", sum)

    Enter a number: 987

    sum of digits: 9
    sum of digits: 17
    sum of digits: 24
```

1

5. Check whether a number is an Armstrong number

Functions

1. Write a function to find the greatest of three numbers.

```
    def greatest(a , b, c):
        return max(a, b ,c)
    num1 = int(input("Enter first number:"))
    num2 = int(input("Enter second number:"))
    num3 = int(input("Enter third number:"))
    print("Greatest number is:",greatest(num1, num2, num3))

    Enter first number: 11
    Enter second number: 45
    Enter third number: 50

    Greatest number is: 50
```

1

2. Write a function to check whether a number is prime.

```
    def is_prime(n):
        if n <= 1:
            return False
        if n == 2:
            return True
        if n % 2 == 0:
            return False
        for i in range
          if n % i == 0:
            return False
```

1

3. Write a function that returns the reverse of a string.

```
    def is_palindrome(s):
```

```
        reverse = s[::-1]
        if s == reverse:
            return True
        else:
            return False
    word = input ("Enter a string:")
    if is_palindrome(word):
        print(word, " is a palidrome")
    else:
        print(word, "is not a palindrome")

    Enter a string: madam

    madam  is a palidrome
```

1

4. Write a function to calculate the area and perimeter of a rectangle.

5. Write a function that accepts a list and returns the second largest element

excepion handling

1. Write a program to handle ZeroDivisionError.

2. Write a program to handle ValueError when the user enters invalid

```
    try:
        num = int(input("Enter a number:"))
        print("you entered:", num)

    except ValueError:
        print("Invalid input")

    Enter a number: g

    Invalid input
```

1

3. Create a custom exception to check if age is less than 18.

4. Write a program that handles both IndexError and KeyError.

5. Write a program using try, except, else, and finally to read a file and display its contents.